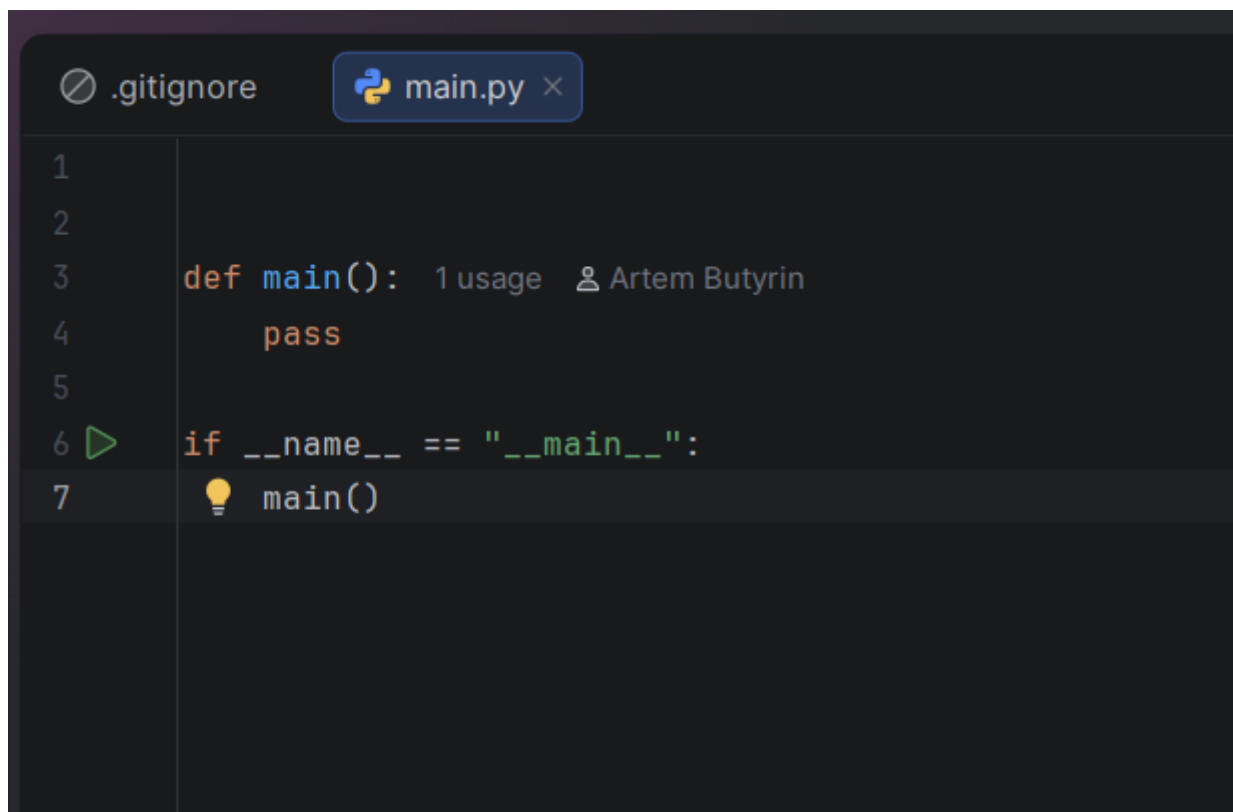
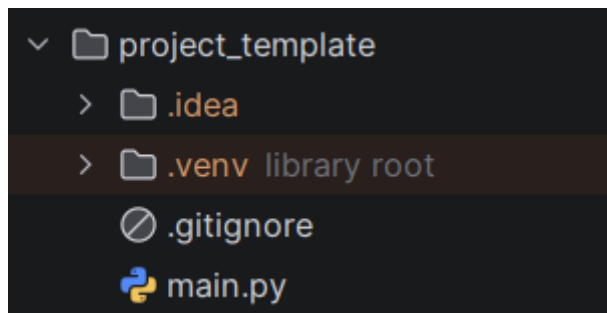


Практична робота №7. Проєкт, модулі, імпорт бібліотек, рір. Робота з файлами у Python.

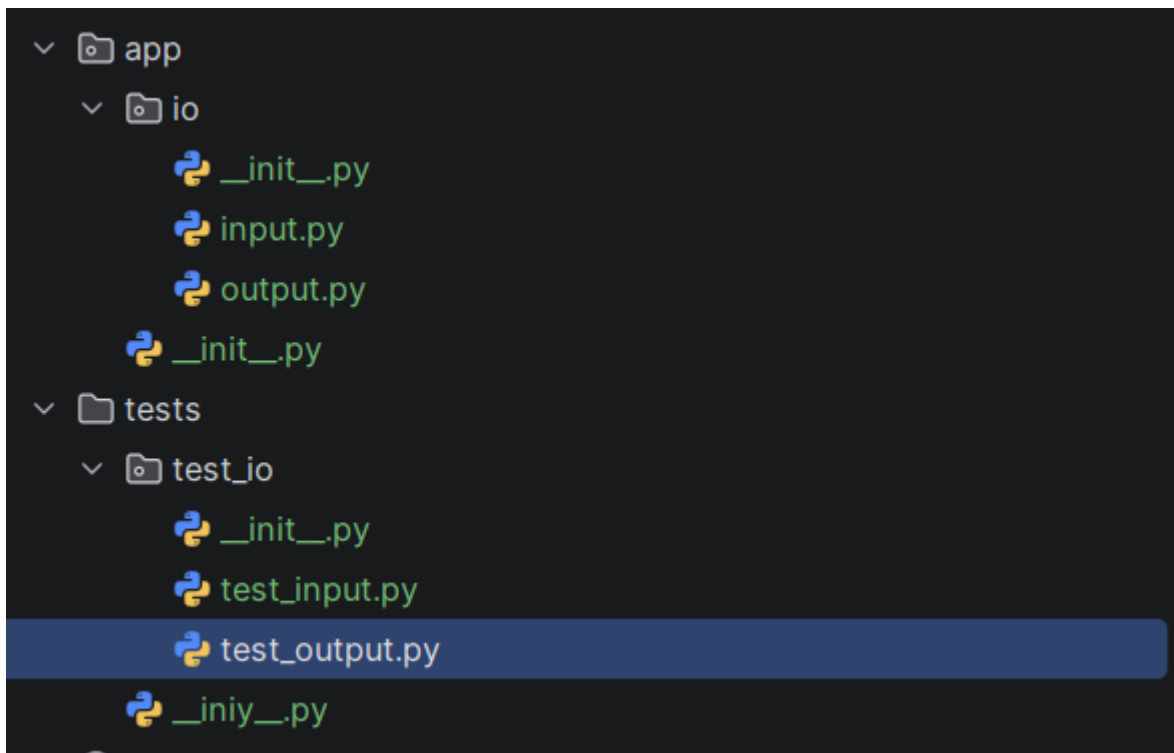
Бутирін Артем (ходжу на 2 групу (за САзом Зтя))

1. Створення нового проєкту.



Репозиторію

2. Структура проєкту.



```
PS C:\Users\Artem\Documents\KMA\Python_Big_Data\pract_7\project_template> git commit -m "commit for HW"
[master 9b9d8e8] commit for HW
13 files changed, 5 deletions(-)
delete mode 100644 pract_7/.idea/.gitignore
create mode 100644 pract_7/live_pract/__init__.py
rename pract_7/{ => live_pract}/local_module/__init__.py (100%)
rename pract_7/{ => live_pract}/local_module/sum.py (100%)
rename pract_7/{ => live_pract}/pract_7_live.py (100%)
create mode 100644 pract_7/project_template/app/__init__.py
create mode 100644 pract_7/project_template/app/io/__init__.py
create mode 100644 pract_7/project_template/app/io/input.py
create mode 100644 pract_7/project_template/app/io/output.py
create mode 100644 pract_7/project_template/tests/__iniy__.py
create mode 100644 pract_7/project_template/tests/test_io/__init__.py
create mode 100644 pract_7/project_template/tests/test_io/test_input.py
create mode 100644 pract_7/project_template/tests/test_io/test_output.py
```

3. Работа з модулями.

```
PS C:\Users\Artem\Documents\KMA\Python_Big_Data\pract_7\project_template> where.exe python
C:\Users\Artem\AppData\Local\Programs\Python\Python311\python.exe
C:\Users\Artem\AppData\Local\Microsoft\WindowsApps\python.exe
```

```
PS C:\Users\Artem\Documents\KMA\Python_Big_Data\pract_7\project_template> .\venv\Scripts\activate
(.venv) PS C:\Users\Artem\Documents\KMA\Python_Big_Data\pract_7\project_template> py -m pip install --upgrade pip
Requirement already satisfied: pip in c:\users\artem\documents\kma\python_big_data\pract_7\project_template\.venv\lib\site-packages (25.1.1)
ERROR: Could not install packages due to an OSError: Could not find a suitable TLS CA certificate bundle, invalid path:
C:\Program Files\PostgreSQL\18\ssl\certs\ca-bundle.crt

WARNING: There was an error checking the latest version of pip.
(.venv) PS C:\Users\Artem\Documents\KMA\Python_Big_Data\pract_7\project_template> py -m pip install --upgrade pip --trusted-host pypi.org --trusted-host files.pythonhosted.org
Requirement already satisfied: pip in c:\users\artem\documents\kma\python_big_data\pract_7\project_template\.venv\lib\site-packages (25.1.1)
Collecting pip
  Downloading pip-26.0.1-py3-none-any.whl.metadata (4.7 kB)
  Downloading pip-26.0.1-py3-none-any.whl (1.8 MB)
    1.8/1.8 MB 2.4 MB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 25.1.1
    Uninstalling pip-25.1.1:
      Successfully uninstalled pip-25.1.1
  Successfully installed pip-26.0.1
(.venv) PS C:\Users\Artem\Documents\KMA\Python_Big_Data\pract_7\project_template> py -m pip --version
pip 26.0.1 from C:\Users\Artem\Documents\KMA\Python_Big_Data\pract_7\project_template\.venv\Lib\site-packages\pip (python 3.11)
(.venv) PS C:\Users\Artem\Documents\KMA\Python_Big_Data\pract_7\project_template>
```

```
(.venv) PS C:\Users\Artem\Documents\KMA\Python_Big_Data\pract_7\project_template> pip install numpy --trusted-host pypi.org --trusted-host files.pythonhosted.org
Collecting numpy
  Using cached numpy-2.4.2-cp311-cp311-win_amd64.whl.metadata (6.6 kB)
Using cached numpy-2.4.2-cp311-cp311-win_amd64.whl (12.6 MB)
Installing collected packages: numpy
Successfully installed numpy-2.4.2
```

```
(.venv) PS C:\Users\Artem\Documents\KMA\Python_Big_Data\pract_7\project_template> pip install "pandas==2.2.3" --trusted-host pypi.org --trusted-host files.pythonhosted.org
Collecting pandas==2.2.3
  Downloading pandas-2.2.3-cp311-cp311-win_amd64.whl.metadata (19 kB)
Requirement already satisfied: numpy>=1.23.2 in .\venv\Lib\site-packages (from pandas==2.2.3) (2.4.2)
Collecting python-dateutil>=2.8.2 (from pandas==2.2.3)
  Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting pytz>=2020.1 (from pandas==2.2.3)
  Downloading pytz-2026.1.post1-py2.py3-none-any.whl.metadata (22 kB)
Collecting tzdata>=2022.7 (from pandas==2.2.3)
  Downloading tzdata-2025.3-py2.py3-none-any.whl.metadata (1.4 kB)
Collecting six>=1.5 (from python-dateutil>=2.8.2->pandas==2.2.3)
  Downloading six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Downloading pandas-2.2.3-cp311-cp311-win_amd64.whl (11.6 MB)
11.6/11.6 MB 22.0 MB/s 0:00:00
Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Downloading pytz-2026.1.post1-py2.py3-none-any.whl (510 kB)
Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Downloading tzdata-2025.3-py2.py3-none-any.whl (348 kB)
Installing collected packages: pytz, tzdata, six, python-dateutil, pandas
Successfully installed pandas-2.2.3 python-dateutil-2.9.0.post0 pytz-2026.1.post1 six-1.17.0 tzdata-2025.3
```

«Тепер пропоную вам встановити самостійно додатково пакети matplotlib та pylint, black.»

Також є, там дуже багато тексту в терміналі, не став все показувати

```
(.venv) PS C:\Users\Artem\Documents\KMA\Python_Big_Data\pract_7\project_template> git commit -m "Add requirements.txt"
[master dd8b4e5] Add requirements.txt
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 pract_7/project_template/requirements.txt
(.venv) PS C:\Users\Artem\Documents\KMA\Python_Big_Data\pract_7\project_template> git push
Enter passphrase for key '/c/Users/Artem/.ssh/id_ed25519':
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
```

4. Робота з файлами.

```
input.py ×  _iniy_.py  _init_.py  test_input.py  test_output.py

1  import pandas as pd
2
3  def read_from_console():  Ⓐ Artem Butyrin *
4      """
5      Reads text input from the console
6
7      Returns:    str: The text entered by the user
8      """
9      text = input("Enter text: ")
10     return text
11
12
13  def read_from_file(filepath):  Ⓐ Artem Butyrin *
14      """
15      Reads text from a file using Python built-in capabilities
16
17      Args:    filepath (str): Path to the file to read
18
19      Returns:    str: The content of the file
20      """
21      with open(filepath, "r", encoding="utf-8") as f:
22          return f.read()
23
```

```
24
25  def read_from_file_pandas(filepath):  Ⓐ Artem Butyrin *
26      """
27      Reads data from a file using the pandas library
28
29      Args:    filepath (str): Path to the CSV file to read
30
31      Returns:    DataFrame: The content of the file as a pandas DataFrame
32      """
33      ⚡ return pd.read_csv(filepath)
```

Потім далі

```
input.py  output.py ×  main.py  __iniy__.py  __init__.py

def print_to_console(text):  Artem Butyrin *
    """
    Prints text to the console

    Args:  text (str): The text to print
    """
    print(text)

def write_to_file(text, filepath):  Artem Butyrin *
    """
    Writes text to a file using Python's built-in capabilities.

    Args:
        text (str): The text to write
        filepath (str): Path to the file to write to
    """
    with open(filepath, "w", encoding="utf-8") as f:
        f.write(text)
```

Далі main.py

```
output.py  main.py ×  sample.txt  sample.csv  __iniy__.py  __init__.py  test_input.py

import pandas as pd
from app.io.input import read_from_console, read_from_file, read_from_file_pandas
from app.io.output import print_to_console, write_to_file

def main():  1 usage  Artem Butyrin *
    console_text = read_from_console()

    file_text = read_from_file("data/sample.txt")

    pandas_data = read_from_file_pandas("data/sample.csv")
    pandas_text = pandas_data.to_string()

    print_to_console(console_text)
    print_to_console(file_text)
    print_to_console(pandas_text)

    write_to_file(console_text + "\n" + file_text + "\n" + pandas_text, filepath: "data/output.txt")

if __name__ == "__main__":
    main()
```

+ КОМІТ

5. Написання тестів.

```
(.venv) PS C:\Users\Artem\Documents\KMA\Python_Big_Data\pract_7\project_template> pip install pytest --trusted-host pypi.org --trusted-host files.pythonhosted.org
Collecting pytest
  Downloading pytest-9.0.2-py3-none-any.whl.metadata (7.6 kB)
Requirement already satisfied: colorama>=0.4 in .\venv\Lib\site-packages (from pytest) (0.4.6)
Collecting iniconfig>=1.0.1 (from pytest)
  Downloading iniconfig-2.3.0-py3-none-any.whl.metadata (2.5 kB)
Requirement already satisfied: packaging>=22 in .\venv\Lib\site-packages (from pytest) (26.0)
Collecting pluggy<2,>=1.5 (from pytest)
  Downloading pluggy-1.6.0-py3-none-any.whl.metadata (4.8 kB)
Collecting pygments>=2.7.2 (from pytest)
  Using cached pygments-2.19.2-py3-none-any.whl.metadata (2.5 kB)
Downloading pytest-9.0.2-py3-none-any.whl (374 kB)
Downloading pluggy-1.6.0-py3-none-any.whl (20 kB)
Downloading iniconfig-2.3.0-py3-none-any.whl (7.5 kB)
Using cached pygments-2.19.2-py3-none-any.whl (1.2 MB)
Installing collected packages: pygments, pluggy, iniconfig, pytest
Successfully installed iniconfig-2.3.0 pluggy-1.6.0 pygments-2.19.2 pytest-9.0.2
```

Буде pytest

Для того щоб pytest побачив тести створюю ще conftest.py в app
із вмістом

```
```\n
```

```
import sys
```

```
import os
```

```
sys.path.insert(0, os.path.dirname(__file__))
```

```
```\n
```

```
(.venv) PS C:\Users\Artem\Documents\KMA\Python_Big_Data\pract_7\project_template> pytest tests/
===== test session starts =====
platform win32 -- Python 3.11.9, pytest-9.0.2, pluggy-1.6.0
rootdir: C:\Users\Artem\Documents\KMA\Python_Big_Data\pract_7\project_template
collected 6 items

tests\test_io\test_input.py ..... [100%]

===== 6 passed in 0.63s =====
```

3 тестами

```
1 import pytest
2 import os
3 from app.io.input import read_from_file, read_from_file_pandas
4
5
6 # Tests for read_from_file
7 ▶ def test_read_from_file_returns_string(tmp_path):  Ⓜ Artem Butyrin
8
9     f = tmp_path / "test.txt"
10    f.write_text(data="hello world", encoding="utf-8")
11    result = read_from_file(str(f))
12    assert isinstance(result, str)
13
14
15 ▶ def test_read_from_file_correct_content(tmp_path):  Ⓜ Artem Butyrin
16
17     f = tmp_path / "test.txt"
18     f.write_text(data="hello world", encoding="utf-8")
19     result = read_from_file(str(f))
20     assert result == "hello world"
21
22
23 ▶ def test_read_from_file_not_found():  Ⓜ Artem Butyrin
24
25     with pytest.raises(FileNotFoundError):
26         read_from_file("nonexistent_file.txt")
27
28
29 ▶ def test_read_from_file_pandas_returns_dataframe(tmp_path):  Ⓜ Artem Butyrin
30
31     import pandas as pd
32     f = tmp_path / "test.csv"
33     f.write_text(data="name,age\nAlice,25\nBob,30", encoding="utf-8")
34     result = read_from_file_pandas(str(f))
35     assert isinstance(result, pd.DataFrame)
36
37
38 ▶ def test_read_from_file_pandas_correct_columns(tmp_path):  Ⓜ Artem Butyrin
39
40     f = tmp_path / "test.csv"
41     f.write_text(data="name,age\nAlice,25\nBob,30", encoding="utf-8")
42     result = read_from_file_pandas(str(f))
43     assert list(result.columns) == ["name", "age"]
44
45
46 ▶ def test_read_from_file_pandas_correct_rows(tmp_path):  Ⓜ Artem Butyrin
47
48     f = tmp_path / "test.csv"
49     f.write_text(data="name,age\nAlice,25\nBob,30", encoding="utf-8")
50     result = read_from_file_pandas(str(f))
51     assert len(result) == 2
```

6. Висновки.

а. Що зробили?

Я зробив дуже класну річ, створи в .venv віртуальне середовище
налаштував проєкт з папки як це правильно робиться
купа нових бібліотек pip (numpy, pandas, matplotlib, pylint, black)
реалізація функцій виводу вводу
+ тести pytest

б. Що нового дізнались для себе?

Насправді все, все що в пункті а перерахував це все нове, а ще це крок до реально великих проєктів і це цікавить

с. Що було корисним? Що б Ви використали в майбутньому?

requirements.txt для спільної роботи в команді
структура проєкту з пакетами
тести з pytest
знову ж таки все що було в цьому дз дуже корисне і думаю ще буду повертатися до цього

д. Що можна було б покращити нам для студентів в цій роботі?

Все чудово, не бачу проблем

7. Надсилання звіту.

Надсилаю звіт.